

Predicting Iron Losses in Soft Magnetic Materials With Arbitrary Voltage Supply: An Engineering Approach

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Abstract—We propose a new approach for predicting iron losses in soft magnetic materials with any voltage supply, starting from the knowledge of the iron losses with a sinusoidal or pulsewidth modulation supply. The model is based on the separation of the loss contributions due to hysteresis, eddy currents, and excess losses with the two supplies. Since any contribution depends on the voltage supply characteristics, it is possible to find a direct mathematical relationship between the iron loss contribution and the voltage supply characteristics. As a consequence, an iron loss prediction can be obtained with any voltage supply if it does not produce a hysteresis minor loop. The energetic model is based on coefficients that depend on the magnetic material characteristic. We performed an accurate analysis of the model on eight magnetic materials used for electrical machine construction, of different thicknesses and alloy compositions. In this way, we found the main coefficients for a large spread of magnetic materials. As a consequence, our approach can be a useful support for electrical machine designers when the energetic performance of a magnetic material has to be predicted for a voltage supply different from the sinusoidal one.

Index Terms—Electrical machine design, iron losses, soft magnetic materials.

I. INTRODUCTION

PREDICTING iron losses in magnetic materials when a nonsinusoidal voltage supply is applied [in particular for pulsewidth modulation (PWM) inverter supply] is still an open research field. Different approaches have been proposed by the researchers but, unfortunately, they are quite complex, and the obtained results cannot be considered suitable for an immediate use by electrical machine designers [1]–[4].

Even if the proposed approaches and related results can be considered in good agreement with the measured ones, the authors take into account only one magnetic material, and there are no references about the use of the proposed methods with other materials. All the proposed energetic models use coefficients and parameters that are dependent on the chemical and physical characteristics of the considered material; thus, the work on a single material cannot be exhaustive for a practical application of the proposed models.

In the past, the authors have proposed a useful approach for predicting iron losses based on a statistical analysis of different magnetic materials, but it is limited just to the PWM inverter

supply [5]. In this paper, a different approach is proposed together with a suitable model. This model can be applied every time the characteristics of the voltage supply are known. The presented model has been set up on a large number of magnetic materials having different chemical and physical characteristics, so that the reported results can be useful for a direct application by the electrical designers.

II. SEPARATION OF THE IRON LOSS CONTRIBUTIONS

Following the modern researches on the iron losses in magnetic materials by the physicist world, the iron losses can be separated into three main contributions.

Hysteresis Losses: They are due to the steady-state losses of the Weiss domains when a variable magnetic field is applied to the magnetic material. Their contribution can be evaluated by the following equation:

$$P_h = afB_p^x \quad (1)$$

where

B_p peak value of the flux density;
 f frequency;
 x Steinmetz coefficient.

Excess Losses: They are due to the dynamic losses of the Weiss domains when a variable magnetic field is applied to the magnetic material. From a simple point of view, these losses are due to Block walls' discontinuous movements with the production of the Barkhausen jumps. Since the Barkhausen jumps are very fast, they produce eddy currents and related joule losses [6]. Their contribution can be evaluated by the following equation:

$$P_e = ef^{1.5}B_p^{1.5}. \quad (2)$$

Classical Losses: They are due to the presence of eddy currents in all the conductor materials. The classical losses denomination is typical in the physic environments, but these losses correspond to the eddy-current denomination in the engineering environments. For this reason, in the following these losses will be denominated as eddy currents. Obviously, these eddy currents must not be confused with the other ones due to the excess losses. Their contribution can be evaluated by the following equation:

$$P_{ec} = bf^2B_p^2. \quad (3)$$

The eddy currents are strongly influenced by the skin-effect phenomena. As a consequence, a most accurate relation can be used

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for taking into account the skin effect, as shown in the following equation:

$$P_{ec} = bf^{1.5}B_p^2 \frac{\sinh(d\sqrt{f}) - \sin(d\sqrt{f})}{\cosh(d\sqrt{f}) - \cos(d\sqrt{f})}. \quad (4)$$

The coefficients “ a, b, d, e, x ” in the previous relations are dependent on the chemical and physical characteristics of the considered magnetic material. The total iron losses can be written as the addition of the previous contributions, as shown in the following equation:

$$P_{ir, \sin} = afB_p^x + bf^2B_p^2 + ef^{1.5}B_p^{1.5}. \quad (5)$$

If the skin effect is taken into account, using the next equation

$$P_{ir, \sin} = afB_p^x + bf^{1.5}B_p^2 \cdot \frac{\sinh(d\sqrt{f}) - \sin(d\sqrt{f})}{\cosh(d\sqrt{f}) - \cos(d\sqrt{f})} + ef^{1.5}B_p^{1.5}. \quad (6)$$

It is important to remark that the coefficients “ a, b, x, e ” in the two previous relations are not equal *a priori*. The validity of (1)–(6) is restricted to a sinusoidal supply, and the computed losses are expressed in watts per kilogram.

III. IRON LOSS VARIATION AND SUPPLY VOLTAGE WAVEFORM

When the supply voltage is not sinusoidal, a variation of the iron losses has to be expected depending on the typology of the considered waveform. In particular, the variation of the iron loss contributions is strictly linked to the typology of the voltage waveform. Making reference to an ideal inductor without winding losses, the supply voltage can be written as

$$v(t) = NS \frac{dB}{dt} \quad (7)$$

where

N turn number;

S section of the magnetic core.

If the supply voltage $v(t)$ is alternating and the instantaneous voltage value of the first harmonic has the same sign of the instantaneous value of the voltage $v(t)$, the relation between the peak-to-peak value of the flux density B_{pp} and the supply voltage can be written as

$$\int_0^T |v(t)| dt = 2NSB_{pp}. \quad (8)$$

This condition on the voltage waveform assures that no minor loops are present in the main hysteresis loop. Introducing the average rectified value of an alternate voltage V_{av} , the peak value of the flux density can be written as

$$B_p = \frac{V_{av}}{4NS} T = k \frac{V_{av}}{f}. \quad (9)$$

Decomposing the supply voltage in harmonic series, the flux density can be written as

$$B(t) = \frac{1}{2\pi NS} \sum_n \frac{V_{n, \max}}{nf} \sin(2\pi nft + \varphi_n) \quad (10)$$

and the peak value of the generic harmonic $B_{n, \max}$ is equal to

$$B_{n, \max} = \frac{V_{n, \max}}{n} \frac{1}{2\pi f NS} \quad (11)$$

with $V_{n, \max}$ peak value of the n th harmonic.

Eddy-Current Contribution Variation

On the basis of (3) and (11), the eddy-current contribution due to all flux-density harmonics can be written as

$$P_{ec} = \sigma \sum V_{n, \max}^2 = 2\sigma V_{rms}^2 \quad (12)$$

where

V_{rms} voltage rms value;

σ constant coefficient.

In other words, (12) shows that the eddy-current losses depend on the rms value of the supply voltage.

Hysteresis Contribution Variation

Taking into account (1) and (9), the hysteresis contribution can be written as

$$P_h = \zeta V_{av}^x f^{1-x} \quad (13)$$

where ζ is the constant coefficient. Equation (13) shows that the hysteresis losses are depending on the rectified average value of the supply voltage. As previously mentioned, (13) can be used only if the supply voltage does not produce minor loops in the complete hysteresis cycle.

A direct consequence of the previous considerations is that the prediction of the iron losses with arbitrary voltage waveform requires the separation of the iron-loss contribution into their two components. As it will be discussed in the following, the experimental results have shown that the excess losses can be included in the classical losses to define the eddy-current loss term, which are well known by the electrical machine designers.

Relationships Between the Supply Voltage and the Iron Losses

Making reference to a wound core and to the separation of the iron losses contributions with sinusoidal supply, it is possible to predict the iron losses with an arbitrary supply voltage if the voltage characteristics are known. In particular, if the supply voltage is alternating, the following equation can be adopted for the iron losses prediction:

$$P_{ir} = \alpha V_{av}^x + \beta V_{rms}^2 \quad (14)$$

where α and β are constant coefficients. It is important to emphasize that the condition imposed on the voltage waveform is always true for three-phase PWM voltage [4]. As a direct consequence of the imposed conditions on the supply voltage, an arbitrary alternate supply voltage can be characterized by means of the following two parameters:

$$\eta = \frac{V_{av}}{V_{av, fund}} \quad (15)$$

$$\chi = \frac{V_{rms}}{V_{rms, fund}} \quad (16)$$

TABLE I
ITALIAN AISI AND ASTM CODES FOR THE TESTED MATERIALS

Italy	8065	5365	3565	8050	5350	2750	3335	2735
AISI	M46	M43	M19	M47	M45	M15	M36	M19
ASTM	56 F490	64 F360	64 F208	47 F475	47 F305	47 F190	36 F190	36 F158

TABLE II
MAGNETIC AND PHYSICAL CHARACTERISTICS OF THE USED MATERIALS

	Thickness [mm]	Specific losses at 50 Hz - 1.5 T [W/kg]	Specific losses at at 60 Hz- 1.5 T [W/lb]	Minimum flux density [T] at			Anisotropy %	Density Kg/m ³
				2500 A/m	5000 A/m	10000 A/m		
8065	0.65	8	4.70	1.58	1.68	1.78	12	7800
5365	0.65	5.3	3.04	1.52	1.62	1.73	14	7700
3565	0.65	3.5	2.08	1.49	1.60	1.71	14	7600
8050	0.50	8.0	4.50	1.68	1.68	1.78	12	7800
5350	0.50	5.3	3.04	1.54	1.64	1.75	12	7700
2750	0.50	2.7	1.55	1.49	1.60	1.71	18	7600
3335	0.35	3.3	1.90	1.49	1.60	1.71	18	7650
2735	0.35	2.7	1.58	1.49	1.60	1.71	18	7650

therefore, the iron losses with an arbitrary waveform can be rewritten as

$$P_{\text{ir}} = \eta^x P_{h, \sin} + \chi^2 P_{\text{ec}, \sin} \quad (17)$$

where

$P_{h, \sin}$	hysteresis losses with sinusoidal supply;
$P_{\text{ec}, \sin}$	eddy-current losses with sinusoidal supply;
V_{av}	voltage mean rectified value;
V_{rms}	voltage rms value;
$V_{\text{av, fund}}$	mean rectified value of the fundamental voltage;
$V_{\text{rms, fund}}$	rms value of the fundamental voltage.

Equation (17) requires the separation of the iron losses into the two contributions, hysteresis and eddy currents, which usually are not reported on the magnetic material data sheets. If the separation between hysteresis and eddy-current losses and the supply voltage characteristics are known, the iron losses with any kind of voltage supply can be predicted.

IV. SEPARATION OF THE IRON-LOSS CONTRIBUTIONS

The experimental approach adopted for the separation of the iron-loss contribution is reported in the following. By means of a set of experiments, it is possible to obtain the unknown coefficients of (5) and (6) minimizing the following relation:

$$\frac{1}{n} \sum_{i=1}^n \left(\frac{P_{\text{im}} - P^*}{P_{\text{im}}} \right)^2 \quad (18)$$

where

n	number of experimental data;
P_{im}	measured iron losses;
P^*	predicted iron losses by (5) or (6).

The experimental tests have been carried out, on an Epstein frame, with a supply frequency from 10 Hz up to 150 Hz and flux densities from 0.6 T up to 1.7 T. They have been performed

on eight nonoriented magnetic materials usually adopted in electrical machine cores. In particular, three materials with a thickness of 0.65 mm, three materials with a thickness of 0.5 mm, and two materials with a thickness of 0.35 mm have been considered. For each thickness a high, medium, and low-quality material has been considered.

In Table I, the list of the selected materials is reported, using the Italian and the USA codes. In the following, the Italian code will be used for the material identification. It is interesting to specify that the Italian code is composed of four digits. The first two digits show ten times the specific losses in watts per kilogram at 1.5 T, while the second two digits show 100 times the sheet thickness in millimeters.

In order to facilitate the comparison of the used magnetic materials with magnetic materials produced by other companies and present in their data sheets, Table II shows some magnetic and physical characteristics.

In Tables III and IV, the obtained coefficients concerning (5) and (6) are reported for all the eight tested materials. It is interesting to underline that the coefficient “ e ” is always zero for each material and for each used equation. In fact, after a deep analysis, the coefficient “ e ” has been fixed to zero, because the actual value assumed by “ e ” for the eight materials was always very small and in some cases negative. This result does not mean that the excess losses are zero, but that the Epstein frame test does not allow us to differentiate between the eddy currents due to the classical losses and the eddy currents due to the excess losses. As a consequence, for an engineering approach, the effects of the excess losses can be combined with the classical losses to define one global eddy-current loss, as previously mentioned. It follows that the iron losses can be considered as the simple sum of hysteresis and global eddy-current contributions.

In Table V, the average uncertainty band for loss equations (5) and (6) are reported. It is well evident that both the two loss

TABLE III
COEFFICIENTS FOR (5)

Material	a	b	x	e
8065	0.0331	0.000344	1.74	0.00
5365	0.0202	0.000266	1.81	0.00
3565	0.0143	0.000206	1.85	0.00
8050	0.0388	0.000234	1.74	0.00
5350	0.0250	0.000158	1.82	0.00
2750	0.0121	0.000127	1.90	0.00
3335	0.0149	0.000097	2.11	0.00
2735	0.0183	0.000077	1.92	0.00

TABLE IV
COEFFICIENTS FOR (6)

Material	a	b	x	e	d
8065	0.0324	0.00482	1.74	0.00	0.232
5365	0.0197	0.00365	1.81	0.00	0.237
3565	0.0143	0.03807	1.85	0.00	0.016
8050	0.0381	0.00313	1.74	0.00	0.249
5350	0.0250	0.01645	1.82	0.00	0.029
2750	0.0120	0.00201	1.90	0.00	0.197
3335	0.0145	0.00124	2.12	0.00	0.275
2735	0.0181	0.00102	1.92	0.00	0.257

TABLE V
AVERAGE UNCERTAIN BAND IN PERCENTAGE

Material	$\epsilon_m\%$ (Eq.5)	$\epsilon_m\%$ (Eq.6)
8065	$\pm 2.0\%$	$\pm 1.8\%$
5365	$\pm 1.9\%$	$\pm 1.7\%$
3565	$\pm 2.3\%$	$\pm 2.3\%$
8050	$\pm 2.1\%$	$\pm 2.1\%$
5350	$\pm 1.7\%$	$\pm 1.7\%$
2750	$\pm 2.2\%$	$\pm 2.1\%$
3335	$\pm 5.1\%$	$\pm 5.0\%$
2735	$\pm 3.3\%$	$\pm 3.2\%$

equations lead to a good results. As a consequence, the simplest equation, without skin effects [(5), with “e” equal zero], will be considered in the following and the method will be defined as the “*abx* method.”

V. HYSTERESIS AND EDDY-CURRENT CONTRIBUTIONS

The previous results allow us to analyze the weight of the loss contributions for all the considered material. In order to reduce the amount of the presented results, the average values of the coefficients *a*, *b*, *x*, computed on different materials having the same thickness, have been used. As a consequence, only the average percentage contribution of the hysteresis and eddy-current losses, for materials having the same thickness, are reported in the following. Figs. 1–6 show the average contribution of eddy-current and hysteresis losses versus frequency for different

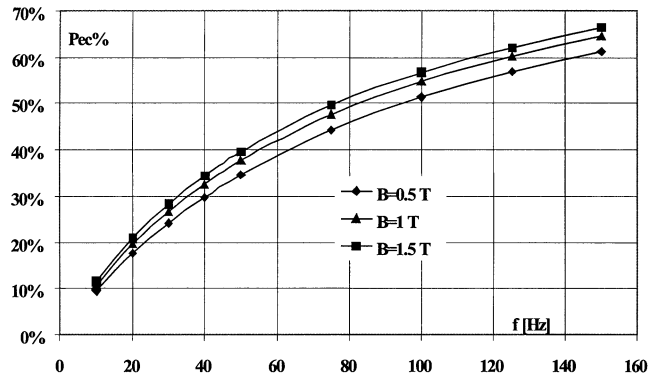


Fig. 1. Average percentage of the eddy-current losses versus frequency for the three materials having thickness 0.65 mm (8065, 5365, 3565).

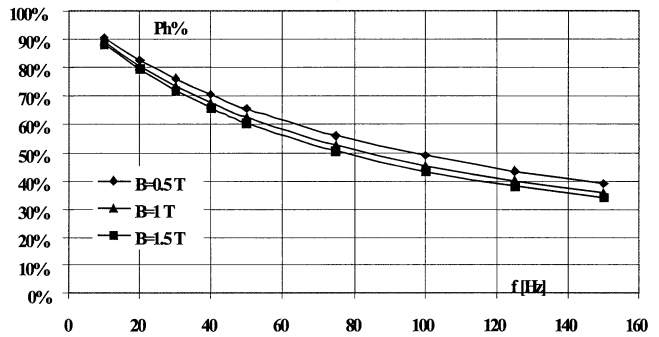


Fig. 2. Average percentage of the hysteresis losses versus frequency for the three materials having thickness 0.65 mm (8065, 5365, 3565).

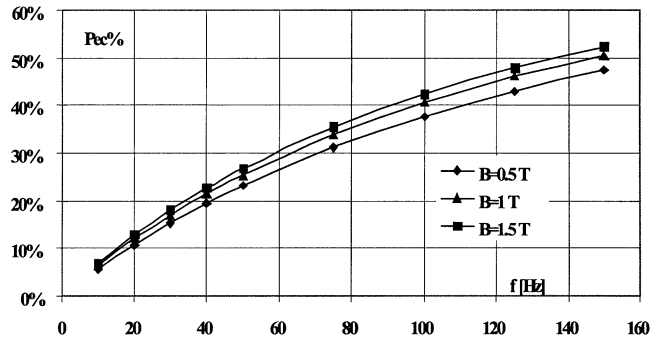


Fig. 3. Average percentage of the eddy-current losses versus frequency for the three materials having thickness 0.5 mm (8050, 5350, 2750).

materials having the same thickness and for three values of flux density. It is interesting to highlight that the separation between hysteresis and eddy-current losses is practically independent with the flux density. In particular, this is well evident in Figs. 5 and 6, where the three curves are overlapped. As a consequence, the percentage of the hysteresis and eddy-current losses can be considered constant with the flux density, and an average percentage can be adopted.

VI. IRON LOSS PREDICTION WITH PWM INVERTER SUPPLY

The methods based on (17) have been tested on all the considered materials for the prediction of the iron losses with PWM inverter supply. The obtained results are reported as a function of the fundamental component of the flux density “*B_{fund}*.” As an

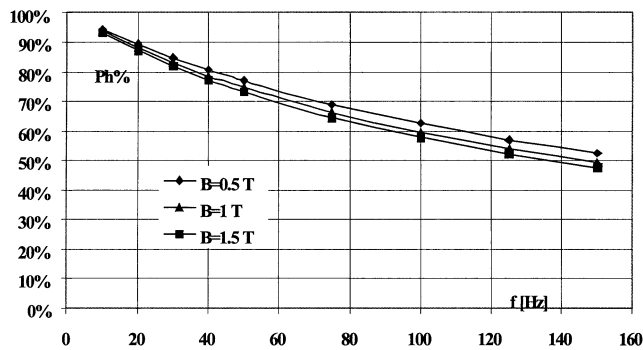


Fig. 4. Average percentage of the hysteresis losses versus frequency for the three materials having thickness 0.5 mm (8050, 5350, 2750).

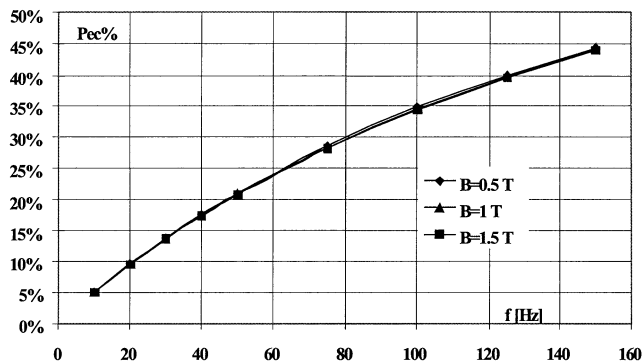


Fig. 5. Average percentage of the eddy-current losses versus frequency for the two materials having thickness 0.35 mm (3335, 2735).

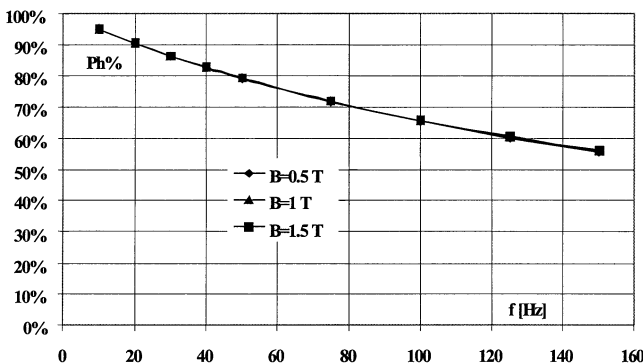


Fig. 6. Average percentage of the hysteresis losses versus frequency for the two materials having thickness 0.35 mm (3335, 2735).

example, Figs. 7–9 show the comparison between the measured and the predicted losses on the 8065, 5350, and 2735 materials when a PWM supply having 1-kHz switching frequency and 0.45 modulation index and a fundamental frequency of 50 Hz is applied. It is well evident the good agreement between the measured and the predicted results. The error of the predicted iron losses with PWM supply is always less than 10% with an average error less than 5%. The average value of the coefficients “ a ,” “ b ,” and “ x ” for the three 0.5-mm tested materials have been used to predict the iron losses on an unknown material having a thickness of 0.5 mm. The comparison between measured and predicted results is reported in Fig. 10. Taking into account that the method has been applied using averaged coefficients, the obtained result can be considered pretty good.

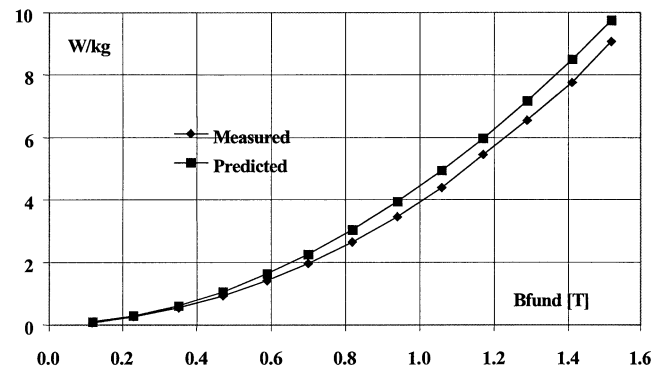


Fig. 7. Comparison between the measured and the predicted losses on the 8065 material with PWM supply versus fundamental flux density (switching frequency = 1 kHz; modulation index = 0.45; fundamental frequency 50 Hz).

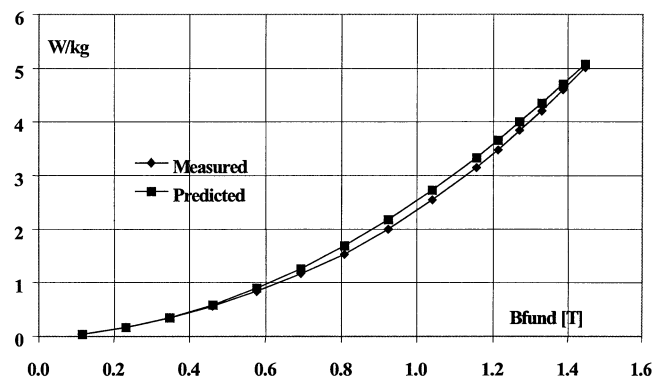


Fig. 8. Comparison between the measured and the predicted losses on the 5350 material with PWM supply versus fundamental flux density (switching frequency = 1 kHz; modulation index = 0.45; fundamental frequency 50 Hz).

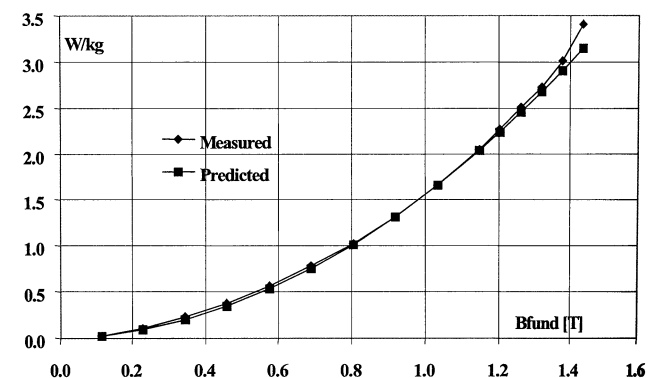


Fig. 9. Comparison between the measured and the predicted losses on the 2735 material with PWM supply versus fundamental flux density (switching frequency = 1 kHz; modulation index = 0.45; fundamental frequency 50 Hz).

VII. ALTERNATIVE METHOD FOR THE PWM IRON LOSS PREDICTION

If the ratio between the carrier frequency and the modulation frequency is high, the η coefficient can be considered equal to one, as verified in a high number of measurements on the three-phase PWM inverter. As previously discussed no minor loops are present in the hysteresis cycle. As a consequence, the iron loss increases with PWM supply are due to an increase of the

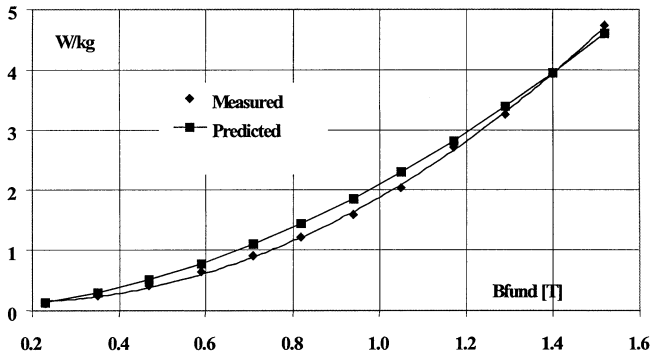


Fig. 10. Comparison between the measured and the predicted losses on an unknown material with thickness of 0.5 mm with PWM supply versus fundamental flux density (switching frequency = 1 kHz; modulation index = 0.8; fundamental frequency 50 Hz).

eddy current only. In particular, in the case of PWM supply, (17) can be rewritten as

$$P_{ir, PWM} = P_{h, sin} + \chi^2 P_{ec, sin} \quad (19)$$

with

$P_{ir, PWM}$ total iron losses with PWM supply;
 $P_{h, sin}$ hysteresis losses with sinusoidal supply;
 $P_{ec, sin}$ eddy-current losses with sinusoidal supply.

It is important to remark that the $P_{h, sin}$ and $P_{ec, sin}$ have to be measured with sinusoidal voltage equal to the PWM fundamental component.

With reference to (17), (19) can be modified as

$$P_{ir, PWM} = P_{ir, sin} + (\chi^2 - 1)P_{ec, sin}. \quad (20)$$

As a consequence, the eddy-current contribution can be directly measured by means of the difference between the iron losses measured with PWM supply and the measured ones with sinusoidal supply. In particular, the eddy-current losses contribution can be computed by the following relation:

$$P_{ec, sin} = \frac{P_{ir, PWM} - P_{ir, sin}}{(\chi^2 - 1)} \quad (21)$$

and a new b' coefficient can be defined by

$$b' = \frac{P_{ec, sin}}{B_{p, fund}^2 f_{fund}^2} = \frac{P_{ir, PWM} - P_{ir, sin}}{(\chi^2 - 1) B_{p, fund}^2 f_{fund}^2} \quad (22)$$

with

$B_{p, fund}$ peak value of the fundamental flux-density component;

f_{fund} frequency of the modulation waveform.

The iron losses with PWM inverter supply can be obtained by the following relation:

$$P_{ir, PWM} = P_{ir, sin} + (\chi^2 - 1) B_{p, fund}^2 b' f_{fund}^2. \quad (23)$$

In the following, this method will be defined as “ b' method.” The new b' coefficient has the same meaning of b coefficient of Table III. In particular, it represents the amount of eddy-current losses in the iron sheets. Since this coefficient is calculated starting from a PWM excitation of the core, it is representative of the skin effect in the iron eddy currents, caused by the high

TABLE VI
AVERAGE VALUE OF b' FOR THE TESTED MATERIALS

Material	b'
8065	0.000207
5365	0.000171
3565	0.000138
8050	0.000154
5350	0.000113
2750	0.000081
3335	0.000071
2735	0.000062

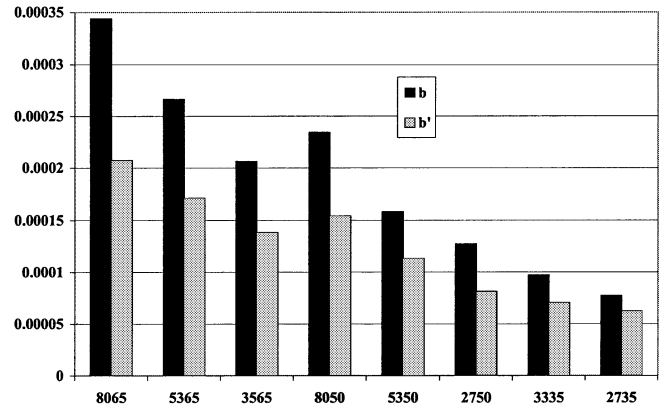


Fig. 11. Comparison between the values of the coefficients b and b' for the materials under tests.

switching frequency of the PWM supply. For this reason, the b' values are significantly lower than the corresponding values of b . The determination of the “ b' ” coefficient has been made using a test with sinusoidal voltage at 50 Hz and several tests performed with a PWM voltage supply at the same fundamental frequency with different modulation indexes and flux-density values. All the tests with PWM voltage have been performed using a switching frequency of 1 kHz. For each test with PWM supply, a “ b' ” value has been obtained. In Table VI, an average value for the considered materials is reported.

The comparison between the values of b (Table III) and b' is reported in Fig. 11 for the materials under test. The computations are referred to a switching frequency of 1 kHz.

In Figs. 12–14, the comparison between the measured and the predicted iron losses, with PWM supply, is reported for three materials of different thickness, when a modulation index of 0.45 and a switching frequency of 1 kHz are used.

VIII. ABOUT THE USE OF THE TWO PROPOSED METHODS

In the previous sections, two methods have been proposed to obtain a relation between the specific iron losses versus the fundamental flux density when the magnetic material is supplied by a PWM voltage waveform. From the obtained results the two methods can be considered equivalent, but are different from the adopted measurement system. The “ abx ” method is based on measurements performed with sinusoidal supply at several frequencies and flux-density values. The “ b' ” method

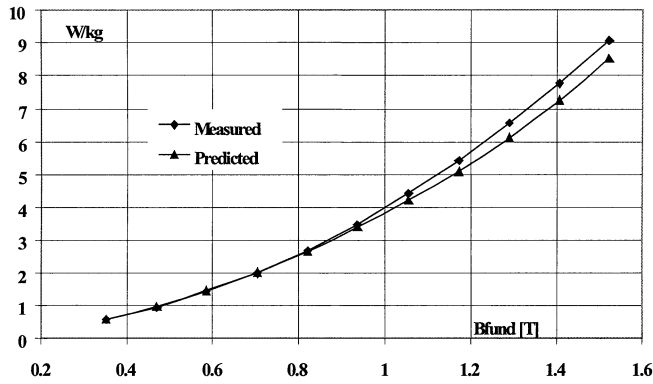


Fig. 12. Comparison between the measured and the predicted PWM iron losses versus fundamental flux density on 8065 material (switching frequency = 1 kHz; modulation index = 0.45; fundamental frequency 50 Hz).

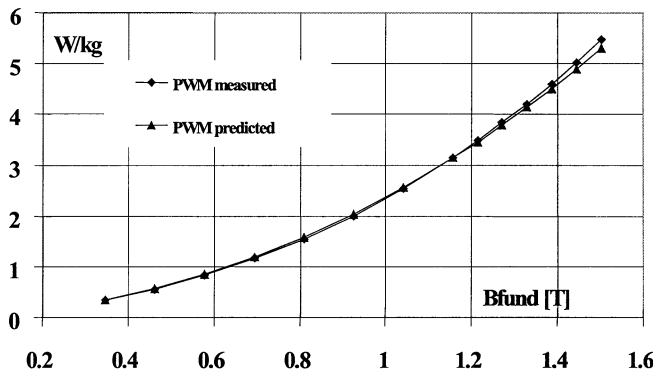


Fig. 13. Comparison between the measured and the predicted PWM iron losses versus fundamental flux density on 5350 material (switching frequency = 1 kHz; modulation index = 0.45; fundamental frequency 50 Hz).

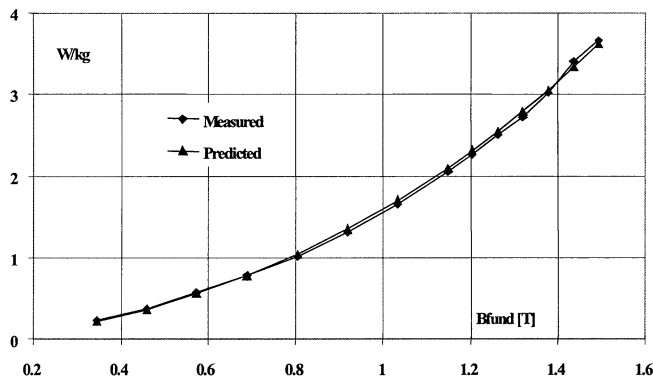


Fig. 14. Comparison between the measured and the predicted PWM iron losses versus fundamental flux density on 2735 material (switching frequency = 1 kHz; modulation index = 0.45; fundamental frequency 50 Hz).

is based on one test with sinusoidal supply and one frequency plus several tests with PWM supply at the same fundamental frequency with different modulation indexes and flux-density values. As previously discussed, the “b’” method allows a separation between the hysteresis and eddy-current contributions taking into account the high-frequency eddy currents due to the PWM supply. Anyway, the accuracy of the two methods is very similar, and the method choice can be made on the basis of the available voltage supply to perform the required tests.

TABLE VII
AVERAGED α FOR THE CONSIDERED MATERIALS AT 50 Hz

	“a, b, x” method			“b’” method		
	0.65 mm	0.5 mm	0.35 mm	0.65 mm	0.5 mm	0.35 mm
α min	0.31	0.21	0.17	0.17	0.13	0.12
α ave	0.38	0.26	0.21	0.23	0.16	0.14
α max	0.44	0.36	0.26	0.26	0.20	0.17

IX. APPLICATION OF THE TWO METHODS FOR THE PWM IRON LOSSES IN ELECTRICAL MACHINES

The most important result obtained by the work is that if the amount of hysteresis or eddy-current losses to the total amount of iron losses is known, it is possible to predict the iron losses with every arbitrary voltage supply taking into account the limits described in the previous sections. Starting from (20) and considering the following ratio:

$$\alpha = \frac{P_{ec, \sin}}{P_{ir, \sin}} \quad (24)$$

the following relation can be obtained:

$$P_{ir, PWM} = [1 + (\chi^2 - 1)\alpha]P_{ir, \sin}. \quad (25)$$

The coefficient α is nearly constant with the flux density, while it is strongly influenced by the fundamental frequency and the material characteristics (thickness, alloy, etc.).

If the magnetic sheet used for the electrical machine core is known, it is not a problem to obtain the parameter α by the coefficients collected in the previous tables. On the contrary, if the magnetic sheet is not well known, an average value of the parameter α can be used. In Table VII, the minimum, average, and maximum values of α are reported for materials having the same thickness and computed by the coefficients obtained by the “abx” and “b’” methods. The values reported in Table VII have been obtained for a fundamental frequency of 50 Hz and they can be used with this fundamental frequency only.

The average α values, obtained by two methods, have been used for the prediction of the iron losses in PWM supplied electrical machines. In particular, a three-phase transformer and three-phase induction motor at synchronous speed have been taken into account. For both the machines the magnetic material characteristics used in their cores were not known, but their thickness was. In Figs. 15 and 16, the predicted iron losses are compared with the measured ones, as a function of the fundamental electromotive force. Both methods give good results with the transformer with an average error better than the 5% between the measured and the predicted results. On the contrary, both methods do not give the same good results on the induction motor. In particular, the “a, b, x” method shows the best performance, with an average error of about 20%÷15%, between the measured and the predicted results.

In order to understand the limits of the proposed methods, it is important to have a critical approach to the obtained results. Obviously, better performances of the methods would be obtained if the magnetic material used in machine cores were known. The two methods give their best results with the three-phase

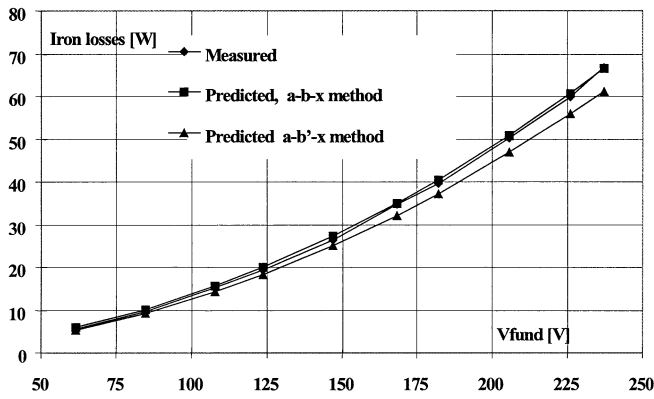


Fig. 15. Measured and predicted iron losses in a three-phase transformer supplied by PWM voltage versus fundamental flux density (switching frequency = 3 kHz; modulation index = 0.7; fundamental frequency 50 Hz).

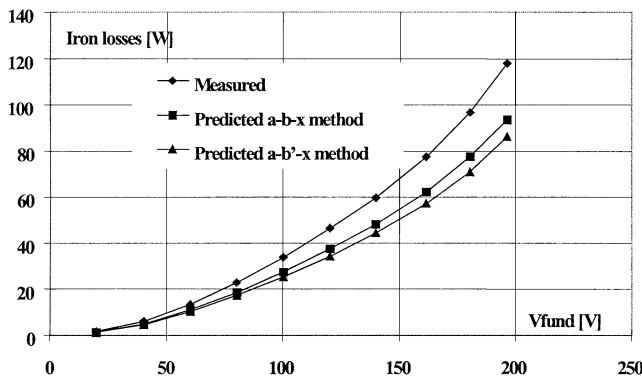


Fig. 16. Measured and predicted iron losses in a three-phase induction motor supplied by PWM voltage versus fundamental flux density (switching frequency = 2.1 kHz; modulation index = 0.7; fundamental frequency 50 Hz).

transformer. This result can be well justified considering that the transformer core has a geometrical shape very close to the Epstein frame shape which was used for every test performed in the present work.

On the contrary, the motor core shape is very different with respect to the Epstein frame. In particular, the mechanical stresses applied to the material during the punch process change the magnetic and energetic characteristics of the material. These effects have been analyzed by the authors in [7], [8], where the deterioration of the magnetic characteristics have been taken into account. In addition, it is important to underline that the term “iron losses” is a conventional term. By an induction motor no-load test at synchronous speed (connecting the motor under test to a synchronous motor with the same poles pair), the iron losses can be conventionally obtained by the following equation:

$$P_{ir} = P_{no-load} - 3R_s I_s^2 \quad (26)$$

with

- P_{ir} iron losses [W];
- $P_{no-load}$ no-load electrical power [W];
- R_s stator resistance [Ω];
- I_s rms no-load current [A].

As direct consequence of (26), all types of losses that are not evaluated as stator joule losses ($3R_s I_s^2$) are automatically defined as iron losses. The impossibility to separate the conven-

tional iron losses [obtained by (26)] into the actual iron losses plus the other non-well-identified loss contributions represents the most important error factor in the comparison between predicted and measured iron losses.

X. CONCLUSION

In this paper, two methods for predicting iron losses with an arbitrary supply have been presented. In particular, the two methods have been set up with a PWM supply. The methods have been tested on eight magnetic materials having different thicknesses.

The considered magnetic materials cover the normally used nonoriented materials in electrical machine production, starting from the cheapest up to the more expensive ones. In the paper, key coefficients for a direct application of the methods by the electrical machine designers are included.

The application of the two methods for the prediction of the PWM losses in the magnetic material itself are very good, and so are the results when the methods are used on transformers (5% 6%).

The methods show higher discrepancy in the prediction of PWM iron losses in induction motor with average errors between 10%–20%. These errors are comparable with the methods described by other authors, but the proposed methods are much more simple and the reported coefficients are of immediate use by the designers, if the thickness of the used magnetic material is known.

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